



TECHNICAL REQUIREMENTS OF AI BASED SMART CLAUSE SELECTION TOOL

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Table of Contents

1. Purpose and scope:	5
2. Control of Document:	5
3. Smart Clause Selection Tool:	5
3.1 Annexure 1: Technical Requirement Document	5
3.1.2. Output: Excel (XLS/XLSM) file generated strictly as per the Buyer's pre-defined RFQ technical template.	5
3.1.3. Functional Requirements:	5
3.2 Annexure 2A: Commercial Terms & Conditions – Indian Bidders	6
3.2.1. Input: User-defined input parameters including but not limited to ..	6
3.2.2. Output:	6
3.2.3. Functional Requirements:	6
3.3 Annexure-2B: Commercial Terms & Conditions – Foreign / Global Tenders	7
3.3.1. Functional Requirements	7
3.4 Annexure-3: Creation of Compliance Format for Purchase Preference Policy – Make in India (MII)	8
3.4.1. Input: User-defined commercial parameters but not limited to:	8
3.4.2. Output:	8
3.4.3. Functional Requirements:	8
3.5 Annexure-4: Creation of Price Bid Format	8
3.5.1. Input	8
3.5.2. Output	8
3.5.3. Functional Requirements	9
3.6 Enclosures: Annexing relevant documents to the tender	9
3.6.1 Input: User-defined input parameters	9
3.6.2. Output: PDF files	9
3.6.3. Functional Requirements:	9
3.7 Contract Clauses: Creation of Contract clauses based on bidder compliance	9
3.7.1. Input:	9
3.7.2. Output:	9
3.7.3. Functional Requirements:	9
4. Output generation based on user parameters and AI:	10
4.1 Description:	10
4.2 RFQ Package Generation:	10



4.3 Contract Clauses:	10
4.4 Complete Output Package Content:	10
4.5 Accuracy Level of the Output:	11
5. Excel output formatting and template compliance requirements:.....	11
5.1 Contract Clauses:	11
5.1.1. Template Compliance:	11
5.1.2. Layout and Formatting:.....	11
5.1.3. Images, Drawings, and Tables:	12
5.1.4. Cell Protection and Editability:	12
5.1.5. Usability and Consistency:	12
6. Dynamic library / master data:	12
6.1 Purpose and Scope:	12
6.2 Dynamic Updates & AI-Based Processing:	13
6.3 Database Management Module:	13
6.4 Integration and Reporting:	13
7. Human-In-Loop and Reinforcement Learning mechanism:	13
8. Steps involved in implementation.....	13
8.1 Indicative System Architecture:	14
8.2 Indicative Technology Stack (Preferred).....	14
9. Deployment, Hosting and Licensing Requirements:	16
10. Milestone - Based Delivery and Installation Plan:.....	16
10.1 Phase 1: Project Initiation & Requirement Finalization	16
10.2 Phase 2: System Design & Architecture	17
10.3 Phase 3: Application Development & Configuration	17
10.4 Phase 4: System Integration and Testing:	17
10.5 Phase 5: User Acceptance Testing (UAT)	18
10.6 Phase 6: Deployment and Installation	18
10.7 Phase 7: Training & Knowledge Transfer	19
10.8 Phase 8: Post Go-Live Support	19
11. SCOPE INCLUSIONS AND EXCLUSIONS.....	19
11.1 Scope Inclusions:	19
11.2 Scope Exclusions:	20
11.3 Support from HAL:.....	21
12. Requirements gathering & analysis:	21
13. System design:	21

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14. Development:	22
15. Testing:	22
16. Training and Technical Support:	22
17. WARRANTY:	22
17.1 Maintenance and Support:	23
18. Replication and Backup	23
19. Hardware Requirements	23
20. List of Deliverables	23
21. Non-Disclosure Agreement (NDA)	24
22. Licensing Perpetual and Unlimited Concurrent Use	24
23. Exchange Agreements	24
24. Malicious Code Certificate	24
25. Acceptance	24
26. Security and Compliance:	25
27. Software Licensing Compliance	25

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1. Purpose and scope:

The purpose and scope of this document is to define the technical requirements of Smart Clause Selection Tool which is a **web-based solution** that is **simple and straightforward to use**, accessible via a web browser and implemented within HAL network.

2. Control of Document:

This document is prepared as part of **Request for Quotation (RFQ)** requirement for **Smart Clause Selection Tool** for MCSRDC, HAL use. The reproduction and disclosure of this document is strictly prohibited.

3. Smart Clause Selection Tool:

Smart Clause selection tool is a software solution intended for end-to-end Tender document preparation and contract clause selection, which encompasses the following:

3.1 Annexure 1: Technical Requirement Document

Technical requirement document is part of Tender bid referred as Annexure-1 which shall clearly and unambiguously define the complete Technical and general requirements of the product or services that buyer intends to procure. The document is usually published in an 'Excel' (XLS or XLSM) or 'PDF form' format wherein the Buyer requirement section is locked and encrypted, the compliance column is open for bidder to indicate compliance.

3.1.1. Input: Technical Requirement Document provided in PDF or Word format by the user.

3.1.2. Output: Excel (XLS/XLSM) file generated strictly as per the Buyer's pre-defined RFQ technical template.

3.1.3. Functional Requirements:

- a. The tool shall automatically parse, extract, and structure technical requirements from unstructured PDF/Word documents using NLP and document intelligence techniques.
- b. The extracted content shall be intelligently mapped to the corresponding fields, rows, and sections of the predefined Excel template. The generated excel output shall ensure:
 - Proper alignment of content within the template structure.
 - Automatic adjustment of row heights and column widths to accommodate the extracted data without truncation or overflow.
 - Preservation of original formatting, including fonts, bulleting, numbering, and indentation to the extent supported by Excel.
 - Images, diagrams, and tables from the source PDF or Word document shall be carried over and represented as-is in the output Excel file, maintaining their original layout, resolution and clarity.

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- Multi-page tables and complex layouts shall be appropriately split or embedded to ensure readability and template compliance.
- c. The output Excel file shall have:
 - Protected (locked) cells for standard headers, clause references, and non-editable sections.
 - Editable cells only where bidder inputs are permitted.
 - The tool shall support regeneration of Annexure-1 in case of updates to the source technical document.

3.2 Annexure 2A: Commercial Terms & Conditions – Indian Bidders

Commercial Terms and Conditions for Indian bidders form an integral part of the Tender bid and are referred to as Annexure-2A. This annexure defines the complete commercial, contractual, and statutory requirements applicable to Indian bidders, including payment terms, delivery conditions, taxes and duties, risk allocation, and other commercial obligations. The document is published in Excel (XLS/XLSX) and PDF formats, wherein standard and mandatory clauses defined by the Buyer are locked, and only permitted fields, are made available for bidder response or confirmation.

3.2.1. Input: User-defined input parameters including but not limited to

- Bidder category (Indian)
- Estimated contract value
- Tender type
- Tender mode
- Category of procurement (Goods, Services, or combined)
- Life-cycle type (Design only, Design and Production, Complete Life Cycle)
- Any other Buyer-defined commercial, statutory, or policy parameters relevant to the tender

3.2.2. Output:

- Excel (XLS/XLSM) file generated strictly as per the Buyer's pre-defined RFQ commercial template for Indian bidders.
- PDF document generated from the finalized commercial terms using the approved Buyer template.

3.2.3. Functional Requirements:

- a. Clause selection shall be performed from a governed **Master Clause Library (Refer Section - 6)**, maintained as master data which is derived from the Purchase Manual, Delegation of Powers, Internal Circulars, Government Circulars, applicable procurement policies and any other document uploaded by the User as a source for master clause library.

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- b. The tool shall automatically select, tailor, and assemble commercial RFQ / Tender clauses based on the configurable parameters including but not limited to parameters selected by the User.
- c. The selected clauses shall be contextually adapted to the RFQ without altering the legal intent or mandatory policy requirements.
- d. The tool shall analyse historical tender and contract data to derive an indicative probability of acceptance for applicable clauses. Such insights shall be presented to the user as decision-support information only, without overriding mandatory policy clauses or Buyer-defined selections.
- e. The generated Excel output shall ensure:
 - Accurate mapping of clauses to the corresponding sections, rows, and fields of the predefined commercial template.
 - Automatic adjustment of row height and column width for readability.
 - Preservation of clause numbering, headings, and formatting standards.
 - Locked (protected) cells for standard terms, headers, and non-editable clauses.
 - Editable cells only where Buyer input or bidder response is permitted.
 - The tool shall generate a professionally formatted PDF version of Annexure-2A reflecting the finalized commercial terms.

3.3 Annexure-2B: Commercial Terms & Conditions – Foreign / Global Tenders

Commercial Terms and Conditions applicable to foreign bidders or Global tenders are referred to as Annexure-2B and form part of the Tender bid documentation wherever applicable. This annexure specifies commercial and contractual provisions relevant to foreign participation, including currency, Incoterms, customs and duties, governing law, and other international procurement considerations. The document is generated in Excel and PDF formats, with Buyer-defined clauses protected from modification and bidder compliance or acceptance indicated only in designated sections

3.3.1. Functional Requirements

Annexure-2B shall be generated by applying the same functional, AI-based clause selection logic, document structure, formatting, and output requirements as Annexure-2A, with clause applicability driven by either (a) Bidder Category set to “Foreign” or “Foreign & Indian” or (b) Type of Tender set to “Global”, thereby enabling automatic inclusion of clauses applicable for foreign procurement terms and conditions, currency, statutory, and policy-specific commercial clauses from Master Clause Library (Refer Section-6).

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3.4 Annexure-3: Creation of Compliance Format for Purchase Preference Policy – Make in India (MII)

Compliance to the Purchase Preference Policy – Make in India (MII) is a mandatory requirement for applicable tenders and is referred to as Annexure-3. This annexure captures bidder declarations and confirmations related to MII eligibility and compliance based on tender value and procurement category. The document is issued in a standard Excel and PDF format, where policy requirements are pre-filled and locked, and bidder compliance is enabled in designated columns.

3.4.1. Input: User-defined commercial parameters but not limited to:

- Estimated Contract Value
- Type of Tender

3.4.2. Output:

- Excel (XLS/XLSM) file generated as per the Buyer's pre-defined and fixed MII compliance output (3 to 4 variations). There is no dynamic change expected in the clauses or text for this output.
- PDF document generated from the finalized MII compliance output.

3.4.3. Functional Requirements:

The tool shall generate the MII compliance format using rule-based logic in accordance with User Input / parameters.

3.5 Annexure-4: Creation of Price Bid Format

The Price Bid Format, referred to as Annexure-4, forms a critical part of the Tender bid and is used by bidders to submit their financial offers. This annexure defines the structure for quoting prices in accordance with the Buyer's requirements, including life-cycle cost elements where applicable. The document is generated in Excel and PDF formats, with all formulae, headers, and standard fields locked, and only designated cells enabled for bidder price inputs to ensure uniformity and accurate evaluation.

3.5.1. Input

- Line Items retrieved from the ERP system or User uploaded .CSV file
- User-defined parameters such as - 'Only Design & Development', 'Design and Production', 'Life-cycle type' so on.

3.5.2. Output

- Excel (XLS/XLSM) file generated strictly as per the Buyer's pre-defined RFQ Price Bid template with in-built formulae.
- PDF document generated from the finalized Price Bid format using the approved Buyer template.

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3.5.3. Functional Requirements

- The tool shall integrate or use a Plug-in to extract line items from the **ERP system** into the predefined Price Bid template.
- The output shall preserve template structure, formatting, and readability, with automatic adjustment of row height and column width.
- Regeneration of Annexure-4 shall be supported in case of changes to ERP data or User parameter(s).

3.6 Enclosures: Annexing relevant documents to the tender

3.6.1 Input: User-defined input parameters

3.6.2. Output: PDF files

3.6.3. Functional Requirements:

- The tool should provide for a document library to index and store all the enclosures.
- The tool shall automatically select relevant enclosures (mostly in PDF format) using the rule-based logic in accordance with User Input / parameters and bundle them together as a single file. The enclosures are pdf / word files such as Bank Guarantee Formats, various statutory declarations, Integrity pact Format etc.
- The tool should provide feature for Admin / Buyer to replace the perform CRUD operations in document library.

3.7 Contract Clauses: Creation of Contract clauses based on bidder compliance

Upon completion of bid evaluation and identification of the L1 bidder, the final contractual terms and conditions are consolidated to form the Draft Contract or Purchase Order. This document is derived by comparing the RFQ clauses with the bidder's compliance submissions and mutually agreed deviations, if any. The output is generated in Word or rich text format and serves as the basis for final contract document, ensuring that mandatory policy, statutory, and Buyer-defined requirements are retained without alteration.

3.7.1. Input:

L1 bidder's compliance document submitted in PDF format, with or without OCR processing

3.7.2. Output:

Word document or rich text format containing the draft contractual clauses

3.7.3. Functional Requirements:

- The tool shall compare RFQ clauses (Annexure-1 document) against the L1 bidder's compliance and identify deviations, acceptances, and conditional responses.
- Bidder responses shall be analysed using Natural Language Processing (NLP) and AI-based techniques to interpret compliance statements.

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- Based on the comparison, the tool shall generate final mutually agreed contractual terms and conditions for inclusion in the Draft Contract or Purchase Order.
- The generated clauses shall be contextually adapted to contract or Purchase Order language without altering legal intent or mandatory policy and statutory requirements.

4. Output generation based on user parameters and AI:

4.1 Description:

The software shall enable users to generate a complete RFQ document package as defined in Sections 3.1 to 3.6, and contract-related terms and conditions as defined in Section 3.7 as a separate Word or rich text output, by leveraging a combination of user-selected parameters, AI/NLP techniques, rule-based logic, and historical Tender/ RFQ data and contracts terms and conditions from the Buyer Organization.

4.2 RFQ Package Generation:

All finalized output files generated under Sections 3.1 to 3.6 shall be automatically bundled into a single compressed ZIP package for download, storage, and distribution.

4.3 Contract Clauses:

The draft contract clauses generated as per Section 3.7 shall be in Word or rich text format in a separate screen and linked to its appropriate RFQ. The tool shall provide export option to load the generated clauses into ERP system (optional) or seamless copy and paste option preserving the font and layout as per Buyer defined Contract template.

4.4 Complete Output Package Content:

The complete output package generated by the system shall include, as applicable to the tender, the following documents:

- Annexure - 1: Technical Requirement Document (XLS / PDF)
- Annexure - 2A: Commercial Terms & Conditions for Indian Bidders (XLS / PDF)
- Annexure - 2B: Commercial Terms & Conditions for Foreign / Global Bidders (XLS/ PDF), where applicable
- Annexure-3: MII Compliance Format (XLS / PDF)
- Annexure-4: Price Bid Format (XLS / PDF)
- Applicable standard RFQ / Tender documents (PDF or Word) selected through the smart clause selection logic
- Clause Selection and Applicability Summary, indicating mandatory, optional, and user-selected clauses (PDF / XLS)
- System-generated RFQ cover document, including list of enclosures.

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- Draft Contract / Purchase Order Clauses (Word / Rich Text) for post-L1 bidder, where applicable

4.5 Accuracy Level of the Output:

The AI/NLP based clause extraction and selection engine shall provide a confidence score for each clause recommendation made for Annexure 2A, Annexure 2B and Draft Contract generation.

The system shall highlight clauses with lower confidence levels and allow the user to review, modify or override the suggested clauses before finalization of the RFQ document package.

The system shall also maintain traceability indicating the source document, policy reference or clause library entry from which the recommendation was derived.

Note: The solution should demonstrate **minimum 90% accuracy** in clause identification during acceptance testing using HAL provided sample documents. The accuracy shall be measured in the following manner:

Accuracy =
$$\frac{\text{Total number of clauses correctly output by the Software Solution}}{\text{Total number of clauses covered in the Tenders / Contract}}$$

The accuracy shall be measured over a minimum of 30 tenders and contract clauses separately.

5. Excel output formatting and template compliance requirements:

5.1 Contract Clauses:

The excel (XLS/XLSM) file shall be the primary system-generated output for Annexure 1, 2A, 2B, 3 & 4. The tool shall generate all excel outputs strictly in accordance with the Buyer-provided standard templates. The following formatting, layout, and usability requirements shall be adhered to:

5.1.1. Template Compliance:

The standard structure, section order, headings, and clause numbering of the Buyer- defined template shall be strictly preserved.

5.1.2. Layout and Formatting:

- Row height and column width shall be automatically adjusted to ensure complete visibility of content without truncation or overflow.
- Fonts, font sizes, text alignment, and numbering styles shall be maintained consistently as defined in the standard template.
- Line spacing, indentation, and wrapping shall be applied to preserve readability and professional appearance.

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**5.1.3. Images, Drawings, and Tables:**

- Images, drawings, diagrams, and tables from the source documents shall be inserted and aligned correctly with same aspect ratio within the relevant Excel cells or merged cell ranges.
- The original layout, proportions, and clarity of such visual elements shall be preserved to the extent supported by Excel.
- Embedded visuals shall not overlap content or disrupt the template structure.

5.1.4. Cell Protection and Editability:

- Cells containing standard requirements, clauses, headers, policy text, formulas, and reference content shall be locked (protected).
- Cells designated for bidder response or compliance input, typically positioned in the right-hand columns, shall be unlocked and editable.
- Cell protection shall be implemented in a manner that prevents accidental modification of mandatory content while allowing valid bidder inputs.

Note: The cell protection and locking to be applied only after finalisation and approval of the final RFQ template by the User and approving authority.

5.1.5. Usability and Consistency:

- The excel file shall be ready for immediate use (publishing the tender) without requiring manual formatting or corrections.
- All sheets within the workbook shall follow a consistent formatting and protection scheme.

6. Dynamic library / master data:

The tool shall include a **Dynamic Library / Master Data Management** module, serving as a centralized repository for all standard clauses, reference documents, and enclosures required for RFQs, commercial terms, and draft contracts.

6.1 Purpose and Scope:

- a. Maintain a structured **database of clauses** for:
 - Annexure 2A and 2B (commercial terms)
 - Draft contract clauses
 - All related enclosures that form part of RFQs
- b. Initial population of the database will be based on authoritative sources such as:
 - HAL Purchase Manual
 - Delegation of Power (DoP) documents
 - Defence Procurement Manual
 - Circulars, policy documents, and other applicable guidelines

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- Support for **non-OCR processed documents**, enabling extraction of content once digitized.

6.2 Dynamic Updates & AI-Based Processing:

Enable dynamic updates to the database as users upload new PDF documents. The module shall:

- Process uploaded PDFs using OCR (if necessary) and NLP/AI tools to extract relevant clauses.
- Automatically classify and map extracted clauses to the appropriate sections.

6.3 Database Management Module:

- a. Provide a user-friendly interface to manage the database, including:
 - Adding, editing, or deactivating clauses manually
 - Uploading enclosures and associating them with specific RFQs or clauses
 - Search, filter, and categorize clauses for quick retrieval
 - Bulk import/export of clauses and enclosures
- b. Maintain access controls to ensure that only authorized users can modify master data, while others can view it in read-only mode.

6.4 Integration and Reporting:

Clauses and enclosures stored in the dynamic library shall seamlessly populate Annexures, RFQ templates, and draft contracts.

7. Human-In-Loop and Reinforcement Learning mechanism:

- a. Human-in-Loop should be an integral part of the workflow ensuring the final output is validated by the User (Buyer) before download / publishing.
- b. The system design should include reinforcement learning or feedback learning mechanism of the clause generation model for continuous training of the model to achieve high accuracy.

8. Steps involved in implementation

This section describes the conceptual workflow and logical sequence for development of the Smart Clause Selection Tool and is independent of the project execution lifecycle defined in Sections on Milestones and SDLC activities

- Discuss with procurement officers to other specific needs, regulations and technology requirements
- Create a comprehensive Master Clause library (database)
- Build or integrate on AI Tool that selects clauses based on requirements
- Validation of all clauses along with MCSRDC, HAL procurement policy and regulations

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- Roll out the system across all procurement process with an option to enhance with larger procurement platform i.e., enterprise level of HAL.

8.1 Indicative System Architecture:

The proposed solution shall broadly follow a modular architecture consisting of the following logical components:

1. Document Ingestion Module

Responsible for ingestion of technical documents, policy circulars, manuals and templates in PDF/Word formats.

2. Clause Extraction and Processing Engine

NLP/AI based engine responsible for extracting clauses from source documents and mapping them to structured clause library entries.

3. Master Clause Library / Dynamic Library

Central repository containing approved clauses, categorized by procurement type, bidder category, policy applicability and other parameters.

4. Smart Clause Selection Engine

Rule-based and AI-assisted logic to generate applicable clauses for:

- Annexure 2A (Commercial Terms – Indian Bidder)
- Annexure 2B (Commercial Terms – Foreign Bidder)
- Draft Contract

5. Template Generation Module

Generates RFQ documents and annexures based on HAL-defined templates.

6. Integration Layer

Interface with IFS ERP system for retrieval of relevant procurement data required for Annexure-4 (Price Bid Format).

7. User Interface & Administration Module

For clause approval, version control, configuration management and document generation.

Note: Bidders may propose enhancements to this architecture; however, the proposed solution shall adhere to the above functional architecture

8.2 Indicative Technology Stack (Preferred)

The proposed solution should preferably be developed using widely adopted, industry-standard technologies that are scalable, secure, and maintainable within HAL's IT environment.

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The following technology stack is indicative in nature and vendors may propose equivalent technologies.

1. Application Layer:

Python (FastAPI / Django / Flask)
Java (Spring Boot)
.NET Core

2. AI / NLP Processing Layer:

Python based NLP frameworks
TensorFlow / PyTorch
spaCy / HuggingFace Transformers
Scikit-learn

3. Database Layer:

PostgreSQL
MySQL / MariaDB

4. Document Processing:

Libraries or tools capable of handling:
PDF parsing
Word document processing
OCR (if required)
Examples:
Apache Tika
Tesseract OCR
Python document processing libraries

5. Integration Layer:

REST APIs
Secure web services
Database connectors for IFS ERP

6. Deployment Environment:

The system shall be deployable on:
On-premise server environment
Linux or Windows server
Virtualized infrastructure if required

External cloud dependency shall not be mandatory.

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Note: The above technology stack is indicative and provided for guidance. Bidders may propose equivalent or superior technologies provided they meet the functional, security, and deployment requirements specified in this RFQ.

9. Deployment, Hosting and Licensing Requirements:

- **Place of Work:** The end - to - end execution of work shall be carried out onsite at MCSRDC, HAL by deploying their resources by following HAL security and IT guidelines and policies.
- The software shall be web-based, accessible through a standard web browser without requiring client-side installations.
- The application shall be hosted within the HAL network environment, in compliance with HAL's IT security, infrastructure, and data governance policies.
- No data shall be hosted or processed outside the HAL network unless explicitly approved by HAL.
- The software shall be provided with a perpetual license, granting HAL the right to use the system indefinitely.
- The license shall be floating in nature, allowing concurrent usage by multiple authorized users up to the licensed capacity.
- The licensing model shall support role-based access and scalability within the licensed limits

10. Milestone - Based Delivery and Installation Plan:

The project shall be executed in a phased manner with clearly defined milestones, deliverables, and acceptance criteria.

10.1 Phase 1: Project Initiation & Requirement Finalization

Duration: 2 - 3 Weeks

Key Activities:

- Project kick-off meeting with HAL stakeholders
- Finalization of functional and non-functional requirements
- Review and confirmation of RFQ templates, annexures, and clause sources
- Finalization of deployment architecture within HAL network

Deliverables:

- Approved Requirement Specification Document
- Project Plan and Implementation Schedule

Milestone:

- ✓ Requirement Sign-off by HAL

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10.2 Phase 2: System Design & Architecture

Duration: 2 Weeks

Key Activities:

- Application architecture and database design
- Clause master data model design
- Rule-based logic definition (MII, Price Bid, compliance logic)
- AI/NLP workflow design for clause selection and bidder compliance analysis

Deliverables:

- System Architecture Document
- Data Model and Integration Design (IFS ERP – CSV interface)

Milestone:

- ✓ Design Approval submission from Vendor

10.3 Phase 3: Application Development & Configuration

Duration: 8–10 Weeks

Key Activities:

- Development of web-based application modules
- Implementation of:
 - a. Annexure-1 to Annexure-4 generation
 - b. Smart clause selection engine
 - c. Rule-based MII logic
 - d. IFS ERP CSV integration
 - e. Configuration management & version control
 - f. Clause library setup using policy source documents
 - g. Role-based access control

Deliverables:

- Configured application modules
- Clause master data repository

Milestone:

- ✓ Completion of Development Phase

10.4 Phase 4: System Integration and Testing:

Duration: 3 Weeks

Key Activities:

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- Integration testing across modules
- Validation of Excel and PDF outputs
- Security and access testing within HAL environment
- Performance and usability testing

Deliverables:

- Tested application ready for UAT
- Test reports and issue resolution logs

Milestone:

- ✓ System Test Sign-off

10.5 Phase 5: User Acceptance Testing (UAT)

Duration: 2–3 Weeks

Key Activities:

- Execution of UAT scenarios by HAL users
- Run pilot test with RFQs to assess the performance of the system
- Validation of RFQ package generation and contract clause creation
- Resolution of UAT issues and refinements

Deliverables:

- UAT Completion Report by Vendor

Milestone:

- ✓ UAT Sign-off by HAL

10.6 Phase 6: Deployment and Installation

Duration: 1–2 Weeks

Key Activities:

- Deployment of application within HAL network environment
- Configuration of user roles and access
- Data migration (if applicable)
- Final validation in production environment

Deliverables:

- Production-ready system
- Deployment and configuration documentation

Milestone:

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✓ Go-Live Approval

10.7 Phase 7: Training & Knowledge Transfer

Duration: 1 Week

Key Activities:

- End-user training sessions
- Administrator training
- Handover of user manuals and system documentation

Deliverables:

- Training completion report
- User and administrator manuals

Milestone:

✓ Training Completion

10.8 Phase 8: Post Go-Live Support

Duration: 03 Months from the date of completion of training

Activities:

- Issue resolution and stabilization support
- Performance monitoring
- Minor enhancements and fine-tuning

Deliverables:

- Support logs and SLA compliance reports

Milestone:

- ✓ Project Completion
- ✓ Full Tech Stack of the software

11. SCOPE INCLUSIONS AND EXCLUSIONS

11.1 Scope Inclusions:

The scope of this project shall include, but not be limited to, the following:

- a. Design, development, configuration, testing, and deployment of a **web-based AI-based Smart Clause Selection Tool** hosted within the HAL network environment.
- b. Provisioning, installation, configuration, and maintenance of all **required databases** for application operation, including schema design, performance tuning, and backups.

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- c. Provisioning and licensing of any third-party software, frameworks, libraries, or tools required for:
 - AI / NLP processing
 - Document parsing and conversion
 - Excel and PDF generation
 - OCR (where applicable)
 - Version control and audit logging
- d. Generation of complete RFQ documentation including:
 - Annexure-1: Technical Requirement Document
 - Annexure-2A / 2B: Commercial Terms & Conditions
 - Annexure-3: MII Compliance Format
 - Annexure-4: Price Bid Format
- e. AI-based smart clause selection and NLP processing for RFQ and contract clause generation, as defined in the SoW.
- f. Rule-based logic implementation for policy-driven outputs (e.g., MII compliance).
- g. Creation and maintenance of a **governed master clause library** sourced from Purchase Manual, Delegation of Powers, internal and Government circulars, procurement policies, and Buyer-uploaded documents.
- h. Parameter-driven classification of clauses as **Mandatory** or **Optional**, with user-assisted clause selection where permitted.
- i. Automatic bundling of applicable RFQ documents into a **compressed ZIP package** with defined naming conventions and folder structure.
- j. Integration with **IFS ERP system or via CSV upload** for price bid data extraction.
- k. Version control, audit trail, and document regeneration capabilities as necessary.
- l. Post-bid analysis and generation of draft contract clauses based on L1 bidder compliance.
- m. User Acceptance Testing (UAT) support, training, documentation, and knowledge transfer.
- n. Post go-live support and maintenance as per the defined SLA.

11.2 Scope Exclusions:

The following items are explicitly excluded from the scope unless specifically agreed in writing:

- a. Procurement or provisioning of hardware, servers, network infrastructure, or operating systems within the HAL data centre.
- b. Any modification to the core functionality of the IFS ERP system beyond the defined CSV-based data exchange.
- c. Legal vetting, legal opinion, or statutory approval of generated RFQ documents, clauses, or contracts.
- d. Creation, amendment, or interpretation of procurement policies, manuals, or Government circulars.
- e. Commercial negotiations, bid evaluation, or vendor selection decisions.

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- f. Support to external bidders or third parties outside the HAL organization.
- g. Integration with external systems not explicitly mentioned in this Scope of Work.
- h. Any features, reports, or enhancements not explicitly defined in this Scope of Work.

11.3 Support from HAL:

- a. HAL can provide up to 05 development workstations (refer section 19.a of this document) and necessary access within the HAL network for the deployed resources during development of the software solution. The development workstations are air gapped from Internet and are connected to HAL network only.
- b. HAL shall provide functional inputs and domain knowledge required for creation and validation of the Master Clause Library.
- c. HAL shall provide relevant historical tender and contract data required for training and tuning of the AI model.
- d. HAL shall provide approved templates, policy documents, manuals, circulars, and standard formats required for system configuration.
- e. HAL shall nominate responsible personnel for requirement clarification, reviews, testing support, and UAT sign-off.
- f. Internet access on a separate computer can be provided for the deployed team for research and communication purpose. The internet computer shall be used in strict adherence to HAL IT policy and under the supervision of User dept.

12. Requirements gathering & analysis:

The tool development shall begin with a detailed Requirements Gathering & Analysis phase to ensure all functional and non-functional requirements are clearly understood and documented. This phase includes:

- Interacting with key stakeholders to capture requirements for RFQ creation, Annexures, Draft Contracts, and the Dynamic Library / Master Data module.
- Reviewing existing policy documents, manuals, circulars, and other reference material to identify clauses, templates, and rules.
- Preparing a Requirements Specification Document (RSD) covering all business rules, workflows, and compliance requirements.
- Conducting feasibility analysis to assess technical and operational constraints.

13. System design:

Based on the approved requirements, a comprehensive System Design shall be developed to define the architecture and workflows of the application:

- Creation of System Architecture diagrams, illustrating module interactions, databases, and external interfaces.

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- Designing UI/UX mock-ups for modules such as RFQ creation, Annexures management, and the Dynamic Library.
- Defining data models, including relational schemas for clauses, enclosures, and master data.
- Documenting technical specifications for integrations, AI/NLP processing, and reporting modules.

14. Development:

The Development phase shall involve coding and implementing all system modules as per the approved design:

- Development of modules for RFQ generation, Annexure population, Draft Contract management, and Dynamic Library / Master Data.
- Implementation of AI/NLP functionality for clause extraction from uploaded PDFs.
- Integration with version control, security, and role-based access systems.
- Following coding standards and best practices for maintainability and scalability.

15. Testing:

A structured testing phase shall ensure the application is functional, reliable and compliant:

- Unit testing of individual modules to verify correct behaviour.
- Integration testing to ensure seamless interaction between RFQ, Annexures, Draft Contracts, and Dynamic Library modules.
- User Acceptance Testing (UAT) with HAL stakeholders to validate workflows and functionality.
- Performance, security, and regression testing to ensure system robustness.

16. Training and Technical Support:

- One-week training shall be provided onsite in customer place (MCSRDC, HAL) for complete familiarization and demonstration of the features and capabilities of the tool including hands-on training session(s) at no additional cost.
- Onsite support to configure and install the tool at MCSRDC at no additional cost.
- Vendor shall provide **technical support** for resolving the issues faced while using the software solution at MCSRDC at no additional cost.

17. WARRANTY:

The entire software solution including application modules, AI components, integrations and databases shall be covered under a comprehensive warranty for a period of 12 months after go live support (3 months)

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Note: go-live support shall be for a period of 3 months after installation, training and acceptance..

17.1 Maintenance and Support:

Post-deployment, the tool shall be supported through a structured Maintenance and Support phase to ensure system stability and performance:

- Providing bug fixes, patch updates, and minor enhancements at no additional cost.
- Assisting MCSRDC, HAL in onboarding new users or modifying master data as per policy changes at no additional cost.
- Offering technical support for a minimum period of 01 year for any operational or integration issues at no additional cost.
- Periodic review of system performance and suggestions for future enhancements during the technical support period.

18. Replication and Backup

- a. The system shall support **data replication and automated backups** to ensure availability, data integrity and recovery.
- b. All application data, master clause libraries, configuration settings, and audit logs (at table and schema level) shall be included.
- c. Backup frequency, retention and restoration procedures shall comply with Buyer IT and security policies and support recovery without loss of historical or version data.

19. Hardware Requirements

- a. **Development Workstation(s):** For the deployed resources HAL will provide workstation for development of the Software at HAL premises with following specifications:
 - Workstation Model: HP Z4G5
 - Processor: Intel Xeon W-Series with 12 Core, base frequency 3.5 GHz. The frequency is expandable up to 4.7 GHz
 - RAM: 64GB, Type: DDR5 ECC SDRAM, Speed: 4800 MT/s
 - Hard disk: 1 TB SSD
 - Graphics Card: NVIDIA RTX 4000 ADA generation with 16 GB RAM
- b. **Production Server:** The bidder shall specify the hardware and infrastructure requirements necessary to deploy and run the proposed solution within the HAL network, including server, storage, database, network, OS/platform prerequisites, scalability, and high availability needs.

20. List of Deliverables

The tool for final deployment shall be provided in an electronic media (USB) after completion of all the milestones. The source code of the software shall also be included in the deliverable.

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21. Non-Disclosure Agreement (NDA)

The successful bidder shall be required to sign a Non-Disclosure Agreement (NDA) with HAL after placement of the order. The NDA is mandatory as the execution of this project involves access to and handling of confidential documents, data, policies, and contractual information belonging to HAL.

22. Licensing Perpetual and Unlimited Concurrent Use

The software shall be provided to HAL on a perpetual license basis with unlimited concurrent user access within the HAL network environment. The license shall permit unrestricted usage by authorized HAL users without dependency on subscription, time limits, or user-based restrictions

23. Exchange Agreements

- a. The Bidder/Contractor/Supplier shall not store, copy or disclose HAL's data and information whether in written, machine readable or other tangible form or disclosed orally, that is of value to HAL. Also, Bidder/Contractor/Supplier agrees to use HAL's data only for the purpose under this Tender/Contract/Agreement, restrict disclosure to any other person or entity or third party and shall safeguard the provided data including from cyber security risks etc.
- b. Non-adherence to this confidentiality clause by the Bidder/Contractor/Supplier shall be treated, as a material breach of this Tender/Contract/Agreement.
- c. The provisions of this clause shall survive and remain in force notwithstanding the Termination or expiry of this Contract/Agreement.

24. Malicious Code Certificate

The seller should upload following certificate in the bid:

- a. This is to certify that the Hardware/ Software being offered, as part of the contract, does not contain Embedded Malicious code that would activate procedures to:
 - Inhibit the desires and designed function of the equipment.
 - Cause physical damage to the user or equipment during the exploitation.
 - Tap information resident or transient in the equipment/network.
- b. The firm shall be considered to be in breach of the procurement contract, in case physical damage, loss of information or infringements related to copyright and Intellectual Property Right (IPRs) are caused due to activation of any such malicious code in the software.

25. Acceptance

- a. The vendor shall deliver and install the tool in MCSRDC.

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b. MCSRDC will accept the product after completion of training

26. Security and Compliance:

- The Smart Clause Selection Tool shall be deployed as a secure, on-premises enterprise application operating entirely within the HAL intranet environment.
- All the tech stack should be approved by the HAL team before use and deployed in HAL network

27. Software Licensing Compliance

- i) The vendor shall ensure that all software tools, libraries, frameworks and databases deployed solution are properly licensed for commercial usage to HAL.
- ii) The vendor shall provide documentary evidence of valid licenses in the name of HAL if deployed at HAL.
- iii) In case open-source software components are used, the vendor shall ensure that such components are licensed for commercial deployment at HAL, and shall provide a list of such components along with applicable license terms.
- iv) Vendor shall be liable for any legal or financial implications arising out of the use of unlicensed or improperly licensed software components by the vendor.

28. DOCUMENTATION

The following documents in soft copy shall be delivered to HAL before go-live:

- i) User Manual:** Detailed instructions on how end-users can effectively operate and navigate the system. It must include step-by-step guides, screenshots, FAQs.
- ii) Administrator Guides:** The document should cover installation procedures, configuration settings, maintenance protocols, security measures and backup plan and strategies. Additionally, it should outline user management processes, including roles and permissions.
- iii) Architecture Documents:** This section must include a high-level overview of the system architecture, detailing components, data flow diagrams, integration points, system Requirements and technology stack. It should also describe scalability considerations and performance benchmarks.

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